STAT 341: Tutorial Test 1 - Sample 0

First name ....

Last name ....

Signature ....

ID Number ....

Email ....

Time allowed: 50 minutes

Special instructions

- All numeric answers should be written either as a number to three decimal places or as a fraction reduced to its lowest terms.
- If working is not specifically requested, a correct answer will receive full marks. However, an incorrect answer could still receive credit if working is provided. An incorrect answer with no working will receive zero marks.

1. **(6 marks)** The population standard deviation is $\sigma_y = \sqrt{\sum_{u \in P}(y_u - \overline{y})^2/N}$

   (a) *(2 marks)* Show that the population standard deviation is location invariant.

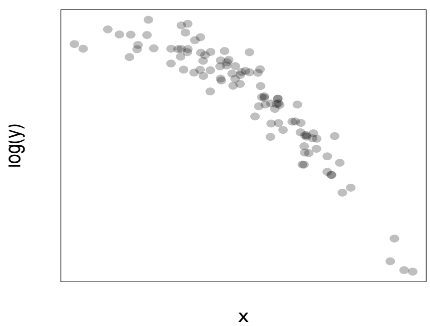
   (b) *(2 marks)* Show that the population standard deviation is scale equivariant.

   (c) *(1 mark)* Show that population coefficient of variation, which is the population standard deviation divided by the average, is scale invariant.

   (d) *(1 mark)* Give an advantage of the coefficient of variation over the population standard deviation.

2. **(5 marks)** Power transformations.

(a) *(2 marks)* Consider the scatterplot below:



Which of the following transformations might straighten this scatterplot?

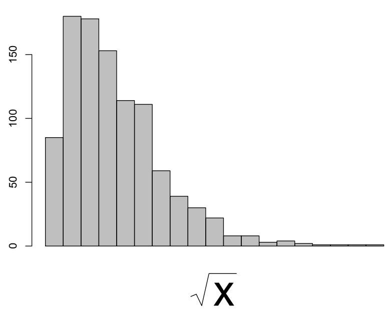
Circle those that **do apply**.

Cross out those that **do not**.

(i) $log(y)$ vs $log(x)$ (ii) y^2 vs $log(x)$

(iii) $log(y)$ vs x^2 (iv) y vs $\sqrt{x}$

(b) *(2 marks)* Consider the histogram below:



Which of the following transformations might make this histogram more symmetric?

Circle those that **do apply**.

Cross out those that **do not**.

(i) $log(x)$ (ii) $-\frac{1}{x}$

(iii) x^2 (iv) x

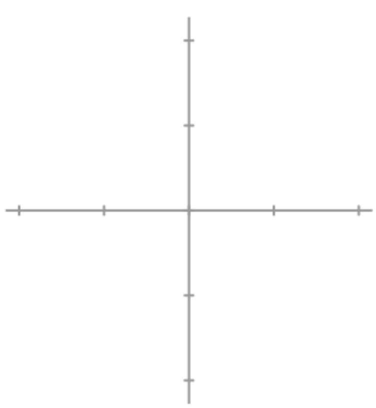
(c) *(1 mark)* Why do power transformations preserve the ranks of the population?

3. **(5 marks)** Consider the location statistic:

$$T_N(y_1, \ldots, y_N) = \overline{y} = \frac{1}{N}\sum_{i=1}^{N} y_i$$

(a) *(3 marks)* Derive this statistic's sensitivity curve:

(b) *(1 mark)* Give a sketch of the sensitivity curve:



(c) *(1 mark)* From this curve, what do you conclude about this statistic's resistance to outliers in y?

4. **(4 marks)** Implicitly defined attributes.

   (a) *(1 mark)* Give an example of an implicitly defined population attribute.

   (b) *(3 marks)* In point form, describe the gradient descent algorithm.

5. **[5 marks]** Determine whether the following statements are true or false.

(a) A scatterplot is not an attribute.

i. True
ii. False

(b) The least squares line is replication invariant.

i. True
ii. False

(c) A histogram is an approximation of a quantile plot

i. True
ii. False

(d) It is preferred for a spread attribute to be location-scale invariant.

i. True
ii. False

(e) A skewness attribute (a measure of asymmetry) should be both scale and location invariant.

i. True
ii. False